MCQ Question Bank

Software (Computer Science)

Topics: Introduction to Programming · Flowcharts · Algorithms · System Software · Application Software · Firmware · Machine Language · Assembly Language · High-Level Languages · Stored Program Concept · MS Office Packages

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Total Questions: 157 | Chapters: 11 | With Answer + Explanation

Chapter 1: Introduction to Programming

Q1. Which of the following best defines a computer program?

A) A set of physical components of a computer
B) A sequence of instructions written to perform a specific task
C) An operating system managing hardware resources
D) A device used to input data into a computer

Answer: (B)

Explanation: *A computer program is a set of step-by-step instructions written in a programming language that tells the computer exactly what to do. Hardware refers to physical components, while the OS is system software.*

Q2. Programming is best described as:

A) Designing computer hardware circuits
B) The process of writing, testing, and maintaining code that instructs a computer
C) Installing software on a computer system
D) Connecting peripheral devices to a computer

Answer: (B)

Explanation: *Programming (or coding) is the full lifecycle of creating software: analysing the problem, writing code, testing it for correctness, and maintaining it over time. It is a purely software activity, not a hardware task.*

Q3. Which of the following is the correct sequence in the program development cycle?

A) Coding → Testing → Problem Analysis → Algorithm
B) Problem Analysis → Algorithm → Coding → Testing → Maintenance
C) Algorithm → Coding → Problem Analysis → Testing
D) Testing → Algorithm → Coding → Maintenance

Answer: (B)

Explanation: *The standard program development cycle always starts with understanding the problem first, then planning the solution (algorithm/flowchart), writing the code, testing it, and finally maintaining the deployed program.*

Q4. What is the primary purpose of writing a program?

A) To manufacture computer hardware
B) To automate tasks and solve computational problems
C) To design graphical user interfaces only
D) To manage network connections

Answer: (B)

Explanation: *The core purpose of a program is to automate repetitive tasks or solve problems that would be time-consuming or error-prone if done manually by humans.*

Q5. Which of the following is NOT a programming language?

A) Python
B) Java
C) HTML (for logic programming)
D) Microsoft Excel (as a standalone language)

Answer: (D)

Explanation: *Microsoft Excel is a spreadsheet application, not a programming language. Python and Java are fully-fledged programming languages. HTML is a markup language used for structuring web pages, not general-purpose programming.*

Q6. A programmer writes instructions in a programming language. These instructions are collectively called:

A) Hardware
B) Firmware
C) Source code
D) Binary data

Answer: (C)

Explanation: *Source code is the human-readable set of instructions written by a programmer in a programming language before it is translated into machine code. Binary data is the machine-level representation after translation.*

Q7. Which of the following best describes 'debugging' in programming?

A) Writing new features for a program
B) The process of finding and fixing errors in a program
C) Compiling the source code into machine code
D) Documenting the program for end users

Answer: (B)

Explanation: *Debugging is the systematic process of identifying, locating, and correcting bugs (errors) in a program. The term originated when an actual moth caused a fault in an early computer.*

Q8. A syntax error in a program means:

A) The program produces the wrong output logically
B) The program violates the grammatical rules of the programming language
C) The program runs out of memory during execution
D) The program cannot connect to the internet

Answer: (B)

Explanation: *A syntax error is a violation of the grammar rules of the programming language (e.g., missing a colon or bracket). The compiler/interpreter catches these before or during execution.*

Q9. A logic error in a program means:

A) The program has incorrect grammar in its code
B) The program crashes during compilation
C) The program runs but produces incorrect results
D) The program cannot find required files

Answer: (C)

Explanation: *A logic error does not prevent the program from running, but the program produces wrong results because the algorithm or formula used is logically incorrect. These are the hardest type of error to find.*

Q10. Which of the following is the FIRST step in solving a programming problem?

A) Writing the code
B) Testing the program
C) Understanding and analyzing the problem
D) Compiling the program

Answer: (C)

Explanation: *Before writing a single line of code, a programmer must fully understand the problem, its inputs, expected outputs, and constraints. Jumping to coding without analysis leads to poor solutions.*

Q11. Documentation in programming refers to:

A) Translating source code to machine code
B) Written descriptions and comments that explain what the program does
C) Running the program to check for errors
D) Designing the physical layout of the computer

Answer: (B)

Explanation: *Documentation includes comments within code and external manuals/guides that explain how the program works. Good documentation makes maintenance and collaboration much easier.*

Q12. Which of the following is an example of an Integrated Development Environment (IDE)?

A) Microsoft Word
B) VLC Media Player
C) Visual Studio Code
D) Google Chrome

Answer: (C)

Explanation: *An IDE is a software application that provides tools like a code editor, compiler/interpreter, and debugger in one place. Visual Studio Code, PyCharm, and Eclipse are popular IDEs.*

Q13. What does 'maintenance' in the program development cycle involve?

A) Writing the initial source code
B) Updating and improving the program after deployment
C) Translating the algorithm into code
D) Designing the flowchart for the program

Answer: (B)

Explanation: *Maintenance is the post-deployment phase where bugs discovered in real use are fixed, performance is improved, and new features are added as requirements change. It is often the longest phase.*

Chapter 2: Flowcharts

Q14. A flowchart is best defined as:

A) A table listing all variables used in a program
B) A diagrammatic representation of an algorithm using standard symbols
C) A written description of how a program should function
D) A type of high-level programming language

Answer: (B)

Explanation: *A flowchart uses standardised graphical symbols (ovals, rectangles, diamonds, parallelograms) connected by arrows to visually represent the logic and flow of an algorithm or process.*

Q15. Which flowchart symbol represents a START or END point?

A) Rectangle
B) Diamond
C) Oval (Terminator)
D) Parallelogram

Answer: (C)

Explanation: *The oval or rounded rectangle (terminator) marks the beginning and end of a flowchart. Every flowchart must have exactly one START and one (or more) END terminator(s).*

Q16. In a flowchart, the rectangle (process box) is used to represent:

A) A decision or condition
B) Input or output operations
C) A processing step or computation
D) The start or end of the program

Answer: (C)

Explanation: *The rectangle represents a process — any action, calculation, or assignment statement (e.g., Total = Price × Quantity). It is the most commonly used symbol in flowcharts.*

Q17. The diamond shape in a flowchart represents:

A) A computation or process step
B) A decision point with two or more outcomes
C) The beginning or end of the flowchart
D) An input/output operation

Answer: (B)

Explanation: *The diamond (decision box) contains a yes/no or true/false question. The flowchart branches into different paths depending on the answer, enabling conditional logic and loops.*

Q18. The parallelogram in a flowchart is used for:

A) Processing steps and calculations
B) Decision making
C) Input and output operations
D) Connectors between different parts of the flowchart

Answer: (C)

Explanation: *The parallelogram represents I/O operations — reading data from the user (input) or displaying results on screen (output). Example: 'Read Name' or 'Print Total'.*

Q19. Arrows in a flowchart indicate:

A) Variables and their data types
B) The flow of control and sequence of steps
C) Memory allocation for the program
D) Comments added by the programmer

Answer: (B)

Explanation: *Arrows (flow lines) connect symbols and show the order in which steps are executed. They define the direction of program flow, including branches and loops.*

Q20. Which of the following is an advantage of using flowcharts?

A) They can be directly executed by a computer
B) They provide a clear visual representation making logic easier to understand
C) They reduce the amount of memory a program uses
D) They replace the need for writing any code

Answer: (B)

Explanation: *Flowcharts make program logic easy to understand even for non-programmers. They are excellent planning and communication tools, helping identify logical errors before coding begins.*

Q21. Which of the following is a disadvantage of flowcharts?

A) They cannot represent decision-making logic
B) For large and complex programs, flowcharts become very large and difficult to follow
C) They cannot show the start and end of a program
D) They are language-dependent and cannot be used with multiple languages

Answer: (B)

Explanation: *For large systems with many branches and loops, flowcharts can sprawl across many pages and become confusing. Modifying them is also time-consuming compared to updating pseudocode.*

Q22. A connector symbol (small circle) in a flowchart is used to:

A) Mark a decision branch
B) Indicate an input operation
C) Connect different parts of a flowchart, especially across pages
D) Represent a subroutine call

Answer: (C)

Explanation: *Connectors (small labelled circles) allow a flowchart to continue on another page or jump to another section without drawing a long crossing arrow, keeping the diagram tidy.*

Q23. Which of the following best describes a 'loop' in a flowchart?

A) A straight path from start to end with no decisions
B) A path that repeats a set of steps based on a condition
C) A direct jump from the start symbol to the end symbol
D) A symbol representing data storage

Answer: (B)

Explanation: *A loop is formed when a flowchart's arrow returns to an earlier point based on a decision diamond's condition. The loop continues executing steps until the condition becomes false.*

Q24. What is the correct shape for representing a predefined process (subroutine) in a flowchart?

A) Diamond
B) Parallelogram
C) Rectangle with double vertical lines on each side
D) Oval

Answer: (C)

Explanation: *A rectangle with vertical lines on both sides indicates a predefined process or subroutine — a module that is defined elsewhere. This avoids repeating its internal details in the main flowchart.*

Q25. Which type of flowchart describes the overall flow of a complex system at a high level?

A) Detailed flowchart
B) System flowchart
C) Data flow diagram
D) Pseudocode chart

Answer: (B)

Explanation: *A system flowchart shows the high-level flow of data and processes across an entire system (e.g., inputs, processing stages, outputs), without detailing the internal logic of each module.*

Q26. Which statement about flowcharts is CORRECT?

A) Flowcharts are written in a specific programming language
B) Flowcharts use standardized symbols for universal understanding
C) Flowcharts can only represent linear (no decision) programs

D) Flowcharts are only used in hardware design

Answer: (B)

Explanation: *Flowcharts use internationally standardised symbols (ISO 5807), making them universally understandable regardless of programming language or nationality.*

Q27. A flowchart is drawn BEFORE writing code because:

A) It automatically generates the source code

B) It helps plan and visualize the logic before actual coding begins

C) It is a legal requirement for all software development

D) It tests the performance of the final program

Answer: (B)

Explanation: *Drawing a flowchart first is a planning technique. It helps the programmer visualise the solution, spot logical flaws early, and serve as a blueprint — reducing costly errors during coding.*

Chapter 3: Algorithms

Q28. An algorithm is best defined as:

A) A programming language used to write software

B) A step-by-step procedure for solving a problem in a finite number of steps

C) A graphical representation of program logic

D) A collection of data stored in a database

Answer: (B)

Explanation: *An algorithm is a well-defined, finite sequence of instructions that solves a specific problem. It is language-independent and can be expressed in natural language, pseudocode, or a flowchart.*

Q29. Which of the following is NOT a characteristic of a good algorithm?

A) Finiteness — it must terminate after a finite number of steps

B) Definiteness — each step must be clear and unambiguous

C) Infiniteness — it should run forever to handle all inputs

D) Effectiveness — each step must be executable

Answer: (C)

Explanation: *A valid algorithm MUST terminate (be finite). An algorithm that runs forever (infinite loop) is not a valid algorithm. The five key properties are: input, output, finiteness, definiteness, and effectiveness.*

Q30. The property of an algorithm that states every step must be precisely and clearly defined is called:

A) Finiteness

B) Input

C) Definiteness

D) Generality

Answer: (C)

Explanation: *Definiteness means each instruction must be unambiguous — there should be no room for interpretation. Every step must have a single, clear meaning so the algorithm produces consistent results.*

Q31. 'Finiteness' as a property of an algorithm means:

A) The algorithm must produce at least one output

B) The algorithm must terminate after a finite number of steps
C) The algorithm must accept at least one input
D) The algorithm must use minimal memory

Answer: (B)

Explanation: *Finiteness guarantees that the algorithm will eventually stop. An algorithm that never terminates cannot be considered correct, as it would prevent the computer from doing any other work.*

Q32. Which of the following correctly describes 'pseudocode'?

A) A special type of machine language
B) An informal high-level description of an algorithm using plain language
C) A strict syntax language used to compile programs
D) Binary code written for embedded systems

Answer: (B)

Explanation: *Pseudocode uses English-like statements mixed with programming constructs (IF, WHILE, FOR) to describe an algorithm without strict syntax rules. It is easier to write than code but more precise than plain English.*

Q33. The time complexity of an algorithm refers to:

A) The physical time it takes to type the algorithm
B) How the execution time grows relative to the size of the input
C) The number of programmers needed to implement the algorithm
D) The amount of disk space the algorithm occupies

Answer: (B)

Explanation: *Time complexity measures how an algorithm's running time scales with input size n. It is expressed using Big-O notation (e.g., O(n), O(n²)). Lower complexity means better performance on large inputs.*

Q34. Which sorting algorithm works by repeatedly swapping adjacent elements if they are in the wrong order?

A) Merge Sort
B) Quick Sort
C) Bubble Sort
D) Insertion Sort

Answer: (C)

Explanation: *Bubble Sort repeatedly compares adjacent pairs and swaps them if they are in the wrong order, causing larger values to 'bubble up' to the end. It is simple but inefficient (O(n²)) for large datasets.*

Q35. A binary search algorithm requires that the data must be:

A) Stored in a linked list
B) Already sorted
C) Stored in a stack data structure
D) Very large in size

Answer: (B)

Explanation: *Binary search works by repeatedly halving the search space. It compares the target to the middle element, then searches the left or right half. This only works if the data is already in sorted order.*

Q36. Which of the following is an example of an algorithm in everyday life?

A) The colour of a computer monitor
B) A recipe for cooking a dish

C) The speed of an internet connection

D) The physical weight of a hard drive

Answer: (B)

Explanation: *A cooking recipe is a perfect everyday analogy for an algorithm — it has defined inputs (ingredients), a sequence of steps (instructions), and produces a specific output (the dish).*

Q37. An algorithm that solves a problem by dividing it into smaller sub-problems of the same type uses the strategy called:

A) Greedy approach

B) Brute force

C) Divide and conquer

D) Dynamic programming

Answer: (C)

Explanation: *Divide and conquer splits a problem into smaller sub-problems of the same type, solves each recursively, then combines results. Merge Sort and Quick Sort are classic divide-and-conquer algorithms.*

Q38. The OUTPUT property of an algorithm states that:

A) The algorithm must take at least two inputs

B) At least one output must be produced

C) The algorithm runs infinitely

D) No external data is required

Answer: (B)

Explanation: *Every algorithm must produce at least one result (output). An algorithm that accepts inputs and performs steps but produces nothing useful is considered incomplete.*

Q39. Which of the following is NOT a way to express an algorithm?

A) Natural language

B) Pseudocode

C) Flowchart

D) Random number table

Answer: (D)

Explanation: *Algorithms can be expressed in natural language (English), pseudocode, or flowcharts. A random number table is a statistical tool, not a method for expressing algorithmic logic.*

Q40. What is the key difference between an algorithm and a program?

A) An algorithm is written in machine code; a program is written in assembly language

B) An algorithm is a logical plan for solving a problem; a program is the implementation of that plan in a specific language

C) A program cannot contain loops but an algorithm can

D) An algorithm always runs faster than a program

Answer: (B)

Explanation: *An algorithm is a language-independent logical solution. A program is the concrete implementation of that algorithm in a specific programming language (like Python or Java) that a computer can execute.*

Chapter 4: System Software

Q41. System software is best defined as:

A) Software specifically designed for word processing tasks
B) Software that manages and controls computer hardware, enabling application software to run
C) Software used only by end users for productivity tasks
D) Software that provides antivirus protection

Answer: (B)

Explanation: *System software forms the foundation layer between hardware and application software. It includes the operating system, device drivers, utilities, and language processors. Without it, hardware cannot function.*

Q42. Which of the following is an example of system software?

A) Microsoft Word
B) Adobe Photoshop
C) Windows 11 Operating System
D) Google Chrome browser

Answer: (C)

Explanation: *Windows 11 is an operating system — the most important type of system software. Microsoft Word, Photoshop, and Chrome are all application software used to perform user-level tasks.*

Q43. The primary function of an Operating System (OS) is to:

A) Allow users to browse the internet
B) Manage hardware resources and provide an interface between the user and hardware
C) Create documents and spreadsheets
D) Play multimedia files

Answer: (B)

Explanation: *The OS acts as an intermediary between the user/applications and the hardware. It manages the CPU, memory, storage, and I/O devices, while also providing a user interface (GUI or CLI).*

Q44. Which of the following is NOT a function of an operating system?

A) Memory management
B) Process management
C) Editing photographs and videos
D) File system management

Answer: (C)

Explanation: *Editing photos and videos is a task performed by application software (e.g., Photoshop, Premiere). The OS handles resource management functions: memory, processes, files, I/O devices, and security.*

Q45. A device driver is a type of system software that:

A) Drives the CPU to execute faster
B) Allows the operating system to communicate with hardware devices
C) Provides end users with productivity tools
D) Manages network traffic over the internet

Answer: (B)

Explanation: *A device driver is a small program that translates OS commands into instructions understood by a specific hardware device (e.g., printer, keyboard, graphics card). Without drivers, hardware cannot function.*

Q46. A utility program is a type of system software designed to:

A) Create business presentations
B) Perform maintenance and optimization tasks on the computer
C) Develop new application software
D) Provide internet access to users

Answer: (B)

Explanation: *Utility programs (system tools) help maintain, configure, and optimize a computer. Examples include antivirus software, disk defragmenters, backup tools, and file compression utilities.*

Q47. Which of the following is an example of a utility program?

A) Microsoft Excel
B) Disk Defragmenter
C) Adobe Illustrator
D) VLC Media Player

Answer: (B)

Explanation: *Disk Defragmenter reorganises fragmented data on a hard disk to improve read/write speed. It is a system maintenance utility. Excel, Illustrator, and VLC are all application software.*

Q48. A language processor (compiler/interpreter/assembler) is considered:

A) Application software
B) System software
C) Firmware
D) Embedded software

Answer: (B)

Explanation: *Language processors translate source code into machine code, enabling programs to run. They are part of system software because they support the development environment rather than performing end-user tasks.*

Q49. The kernel of an operating system is:

A) The user interface of the operating system
B) The core part of the OS that manages system resources at the lowest level
C) An application program bundled with the OS
D) The graphical shell displayed to users

Answer: (B)

Explanation: *The kernel is the heart of the OS, always resident in memory. It handles CPU scheduling, memory allocation, device management, and system calls. The GUI/shell sits on top of the kernel.*

Q50. Which of the following best describes a multitasking operating system?

A) An OS that can run only one program at a time
B) An OS that allows multiple programs to run apparently simultaneously
C) An OS designed specifically for mobile phones
D) An OS that only supports a single user

Answer: (B)

Explanation: *In multitasking, the CPU rapidly switches between multiple processes giving the illusion of simultaneous execution. Modern OSes like Windows, macOS, and Linux all support multitasking.*

Q51. File management as a function of the OS includes:

A) Designing the computer's motherboard
B) Creating, deleting, organizing, and controlling access to files on storage devices
C) Rendering graphics for video games

D) Managing network cables and switches

Answer: (B)

Explanation: *The OS file management system organises files in directories, handles reading/writing operations, enforces permissions, and manages storage allocation. This is done through the file system (NTFS, FAT32, ext4, etc.).*

Q52. Which of the following is an example of a mobile operating system?

A) Windows Server

B) Ubuntu Linux Desktop

C) Android

D) macOS Ventura

Answer: (C)

Explanation: *Android (developed by Google) is the world's most widely used mobile OS. Windows Server and Ubuntu are desktop/server OSes, while macOS runs on Apple Mac computers, not mobile devices.*

Q53. What is the role of a bootstrap loader in system software?

A) It allows users to create boot disks for backup

B) It is the initial program that loads the operating system when the computer is powered on

C) It checks for viruses during program execution

D) It manages the allocation of RAM to different applications

Answer: (B)

Explanation: *The bootstrap loader (or bootloader) is a small program stored in ROM that runs when the computer is powered on. It performs the Power-On Self-Test (POST) and then loads the OS from the storage device into RAM.*

Q54. A Real-Time Operating System (RTOS) is mainly used in:

A) Personal computers for home use

B) Systems that require immediate and predictable response to inputs, such as medical devices

C) Social media platforms and web servers

D) Word processing and spreadsheet applications

Answer: (B)

Explanation: *An RTOS guarantees that critical tasks are processed within a strict time deadline. They are used in aircraft control systems, medical monitors, industrial robots, and anti-lock braking systems (ABS).*

Chapter 5: Application Software

Q55. Application software is best defined as:

A) Software that manages hardware components of a computer

B) Software designed to help the user perform specific tasks

C) Software stored permanently on a ROM chip

D) Software that provides low-level control of the processor

Answer: (B)

Explanation: *Application software is end-user software that directly serves a specific purpose, such as word processing, accounting, gaming, or browsing. It runs on top of system software.*

Q56. Which of the following is an example of application software?

A) Linux Kernel

B) BIOS firmware

C) Microsoft Excel

D) Device driver for a printer

Answer: (C)

Explanation: *Microsoft Excel is a spreadsheet application used by end users for data analysis. The Linux kernel, BIOS, and device drivers are all system-level software, not application software.*

Q57. What is the key difference between system software and application software?

A) System software is used by end users; application software manages hardware

B) System software manages hardware and resources; application software helps users perform tasks

C) Application software is faster than system software

D) System software is always free; application software is always paid

Answer: (B)

Explanation: *System software provides the platform (OS, drivers) that hardware needs to work. Application software sits on top and enables users to do productive work like writing documents, editing images, or playing games.*

Q58. A word processor is an example of which type of software?

A) System software

B) Firmware

C) Application software

D) Assembly language

Answer: (C)

Explanation: *A word processor like Microsoft Word or LibreOffice Writer is application software because it directly serves a user task: creating and editing text documents. It requires system software to run.*

Q59. Which of the following is a General-Purpose Application software?

A) A payroll management program for a specific company

B) Microsoft Word (used for various word-processing needs)

C) An operating system like Windows

D) A BIOS chip program

Answer: (B)

Explanation: *General-purpose application software can be used by many different types of users for a wide range of tasks. Microsoft Word suits students, businesses, authors, etc. Payroll software is a specific-purpose application.*

Q60. Custom (bespoke) application software is:

A) Software sold off-the-shelf for a wide range of users

B) Software developed specifically for the unique needs of a particular organization

C) An operating system tailored for embedded systems

D) Free and open-source software available to everyone

Answer: (B)

Explanation: *Bespoke software is built from scratch to meet the exact requirements of one client. Examples include a hospital's patient management system or a bank's internal transaction processing system.*

Q61. Which of the following is an advantage of off-the-shelf application software?

A) It is perfectly tailored to the specific needs of each organization

B) It is cheaper and immediately available compared to custom software
C) It requires no installation or setup by the user
D) It never needs updates or patches

Answer: (B)

Explanation: *Off-the-shelf software (e.g., Microsoft Office) is mass-produced, reducing cost per user. It is available immediately, well-documented, and supported by large developer teams. The trade-off is it may not perfectly fit specific needs.*

Q62. Which of the following is a disadvantage of custom application software?

A) It cannot be modified to meet organizational needs
B) It is generally more expensive and takes longer to develop
C) It is always inferior in quality to off-the-shelf software
D) It cannot be integrated with other software systems

Answer: (B)

Explanation: *Custom software requires a full development cycle (analysis, design, coding, testing) which is costly and time-consuming. However, the result perfectly fits the client's unique requirements.*

Q63. Which of the following is NOT an example of application software?

A) VLC Media Player
B) Adobe Acrobat
C) UNIX Operating System
D) Google Docs

Answer: (C)

Explanation: *UNIX is an operating system — a type of system software. VLC, Adobe Acrobat, and Google Docs are all application software used directly by end users to complete tasks.*

Q64. Spreadsheet software is primarily used for:

A) Drawing architectural plans and diagrams
B) Organizing, analyzing, and manipulating numerical data in rows and columns
C) Managing network connections and protocols
D) Compiling source code into machine code

Answer: (B)

Explanation: *Spreadsheet applications (Excel, Google Sheets) store data in cells arranged in rows and columns. They support mathematical formulas, statistical functions, charts, and data analysis tools.*

Q65. Database management software is used to:

A) Manage and optimize the CPU's processing cycles
B) Store, retrieve, and manage structured collections of data
C) Provide a graphical interface between the user and the OS
D) Create and edit video files

Answer: (B)

Explanation: *Database Management Systems (DBMS) like MySQL, MS Access, or Oracle allow users to create, read, update, and delete structured data using queries. They ensure data integrity, security, and efficient retrieval.*

Q66. Which of the following is an example of educational application software?

A) Windows Defender
B) Disk Cleanup
C) Khan Academy application

D) GNU Compiler Collection

Answer: (C)

Explanation: *Educational software is designed to teach or support learning. Khan Academy, Duolingo, and Mavis Beacon are examples. Windows Defender is a security utility; Disk Cleanup is a system utility; GCC is a compiler.*

Q67. An application software package that combines word processing, spreadsheet, and presentation tools is called:

A) An integrated development environment

B) An office suite

C) A utility program

D) A compiler toolkit

Answer: (B)

Explanation: *An office suite (like Microsoft Office or LibreOffice) bundles multiple productivity applications together. Users benefit from consistent interfaces and seamless data exchange between the component applications.*

Chapter 6: Firmware

Q68. Firmware is best defined as:

A) Software that can be easily installed and uninstalled by the user

B) A type of software permanently stored in hardware, providing low-level control of a device

C) High-level programming language code used for web development

D) Application software pre-installed on a smartphone

Answer: (B)

Explanation: *Firmware occupies the layer between hardware and software. It is embedded in a device's non-volatile memory and provides the instructions needed to control and initialise that specific device.*

Q69. Firmware is typically stored in:

A) RAM (Random Access Memory)

B) Hard Disk Drive

C) ROM (Read-Only Memory) or flash memory

D) Optical discs like CD-ROM

Answer: (C)

Explanation: *Firmware must persist when the device is powered off, so it is stored in non-volatile memory: ROM or flash memory. RAM is volatile (loses data when power is off), making it unsuitable for firmware.*

Q70. Which of the following is the best example of firmware?

A) Microsoft Office Suite

B) BIOS/UEFI on a computer motherboard

C) Google Chrome browser

D) Linux operating system

Answer: (B)

Explanation: *The BIOS (Basic Input/Output System) or its modern replacement UEFI is classic firmware. It is stored on the motherboard's flash chip and runs before the OS, initialising all hardware components.*

Q71. What is the PRIMARY function of the BIOS firmware?

A) To provide users with word processing capabilities
B) To initialize hardware components during startup and load the operating system
C) To connect the computer to the internet wirelessly
D) To compile high-level language programs

Answer: (B)

Explanation: *During the Power-On Self-Test (POST), BIOS checks all hardware components (RAM, CPU, keyboard, storage), initialises them, and then hands over control to the bootloader to load the OS.*

Q72. Which of the following best describes the relationship between firmware and hardware?

A) Firmware is completely independent of hardware and can run on any device
B) Firmware is embedded within and specifically designed to control a particular hardware device
C) Firmware replaces the need for any hardware in modern computers
D) Firmware is written by end users to customize their hardware

Answer: (B)

Explanation: *Firmware is device-specific. The firmware in a router only works for that router model; it cannot be transferred to a printer. It is closely tied to the hardware it controls.*

Q73. Unlike application software, firmware:

A) Can be easily uninstalled by the user from the computer's control panel
B) Is rarely updated and operates at a lower level, directly interfacing with hardware
C) Requires an internet connection to function properly
D) Is written in high-level languages like Python

Answer: (B)

Explanation: *Firmware operates at the hardware interface level, below the OS. It does not appear in the user's applications list and cannot be uninstalled like regular software. Updates require a special flashing process.*

Q74. Firmware found in devices like washing machines, routers, and printers is an example of:

A) General-purpose application software
B) Embedded firmware controlling the device's specific functions
C) System utilities for optimizing performance
D) Open-source operating systems

Answer: (B)

Explanation: *Consumer electronics contain embedded firmware that handles their specific functions. A washing machine's firmware controls wash cycles; a router's firmware manages packet routing and network protocols.*

Q75. Updating firmware (a firmware 'flash') is done to:

A) Increase the physical memory (RAM) of the device
B) Fix bugs, improve performance, or add new features to the device's hardware control
C) Install a new operating system on the device
D) Format and erase the hard drive of the computer

Answer: (B)

Explanation: *Firmware updates (flashing) overwrite the current firmware with a newer version. They can patch security vulnerabilities, fix operational bugs, or unlock new device capabilities without hardware changes.*

Q76. Which of the following statements about firmware is CORRECT?

A) Firmware is the same as the operating system

B) Firmware sits between hardware and software, providing the lowest level of software control
C) Firmware is always visible and accessible through the operating system's file manager
D) Firmware is a type of application software

Answer: (B)

Explanation: *Firmware bridges the gap between raw hardware and higher-level software. The OS sits above firmware; application software sits above the OS. Firmware is not accessible as a regular file through the OS.*

Q77. UEFI is a modern replacement for which firmware?

A) Device drivers
B) BIOS (Basic Input/Output System)
C) The Windows operating system kernel
D) Assembly language programs

Answer: (B)

Explanation: *UEFI (Unified Extensible Firmware Interface) replaces the older BIOS. It offers advantages including support for hard drives larger than 2TB, faster boot times, a graphical interface, and improved security (Secure Boot).*

Q78. Which of the following best distinguishes firmware from system software?

A) System software runs on hardware directly; firmware requires an OS to run
B) Firmware is embedded in hardware and controls the device itself; system software runs on top of hardware to manage it
C) Firmware is always written in high-level languages; system software is in machine code
D) There is no difference; firmware and system software are the same thing

Answer: (B)

Explanation: *Firmware is embedded inside the hardware device itself and runs without an OS. System software (like the OS) runs after firmware has initialised the hardware, providing a managed environment for applications.*

Chapter 7: Machine Language

Q79. Machine language is defined as:

A) A programming language that uses English-like keywords
B) The lowest-level programming language consisting of binary (0s and 1s) instructions directly understood by the CPU
C) A language used to program robots in manufacturing
D) A symbolic language that uses mnemonics to represent instructions

Answer: (B)

Explanation: *Machine language is the only language a CPU natively understands. It consists entirely of binary digits (0 and 1) representing operations and memory addresses. All other languages must eventually be translated to it.*

Q80. Machine language instructions are written in:

A) Decimal (base 10) numbers
B) Hexadecimal (base 16) numbers only
C) Binary (base 2) — a sequence of 0s and 1s
D) Alphabetic characters and symbols

Answer: (C)

Explanation: *At the hardware level, all instructions are binary (base 2). While programmers sometimes use hexadecimal as a shorthand for binary, the CPU itself processes pure binary signals representing 0 (off) and 1 (on).*

Q81. Which of the following is an advantage of machine language?

A) It is easy to read and understand by programmers

B) It executes very fast because the CPU understands it directly without translation

C) Programs written in it are easily portable across different hardware

D) It requires minimal knowledge of computer architecture

Answer: (B)

Explanation: *Since no translation step is needed, machine language programs execute at maximum speed. This is why performance-critical routines in OS kernels and device drivers sometimes use machine/assembly code.*

Q82. Which of the following is a major disadvantage of machine language?

A) It requires a compiler to translate it before execution

B) It is difficult to write, read, and debug because it consists entirely of binary codes

C) It cannot directly control hardware components

D) It runs slower than high-level language programs

Answer: (B)

Explanation: *Writing long sequences of binary (e.g., 10110000 01100001) is extremely tedious and error-prone for humans. A small mistake — one bit — can cause completely different behaviour, making debugging very difficult.*

Q83. Machine language is said to be machine-dependent because:

A) All machines use the same binary instruction set

B) The instruction set differs for different processors; code written for one CPU will not work on a different CPU

C) It can only run on machines connected to the internet

D) It depends on the machine's operating system version

Answer: (B)

Explanation: *Different CPU families (Intel x86, ARM, RISC-V) have different instruction sets. A machine language program written for an Intel CPU is incompatible with an ARM CPU — they use entirely different binary codes.*

Q84. The set of machine language instructions a particular CPU can execute is called its:

A) Operating system interface

B) Instruction Set Architecture (ISA)

C) Programming paradigm

D) Compiler toolchain

Answer: (B)

Explanation: *The ISA defines the complete set of binary instructions a CPU can understand, the number and size of registers, memory addressing modes, and data types. Examples: x86-64 (Intel/AMD), ARM, MIPS.*

Q85. Machine language is also referred to as the:

A) Third generation language (3GL)

B) Second generation language (2GL)

C) First generation language (1GL)

D) Fourth generation language (4GL)

Answer: (C)

Explanation: *Programming languages are classified by generations. 1GL = machine language (binary), 2GL = assembly language (mnemonics), 3GL = high-level languages (Python, C), 4GL = very high-level/domain-specific (SQL).*

Q86. Does machine language require any translation before execution?

A) Yes, it always needs a compiler to translate it

B) Yes, it needs an interpreter to run line by line

C) No, the CPU directly understands and executes machine language

D) Yes, it needs an assembler to convert it to binary

Answer: (C)

Explanation: *Machine language is the native language of the CPU. No translation is needed — the processor directly fetches, decodes, and executes the binary instructions. All other language types need translation to machine code first.*

Q87. An early example of computing that used machine language programming was:

A) Typing commands into the Windows Command Prompt

B) Programming first-generation computers using punched cards or toggle switches

C) Writing code in Python on a modern laptop

D) Using a spreadsheet to perform mathematical calculations

Answer: (B)

Explanation: *Early computers like ENIAC (1940s) were programmed by physically setting switches or feeding punched cards with binary patterns. This direct machine language programming was extremely laborious.*

Q88. Which generation of programming languages does machine language belong to?

A) 2nd generation

B) 3rd generation

C) 4th generation

D) 1st generation

Answer: (D)

Explanation: *The language generation classification: 1GL (machine language) → 2GL (assembly) → 3GL (high-level: C, Java) → 4GL (SQL, report generators) → 5GL (AI/logic-based: Prolog). Machine language is the first and lowest generation.*

Q89. Portability is a problem with machine language because:

A) It requires a large hard disk to store programs

B) Programs must be completely rewritten when moved to a computer with a different CPU architecture

C) It cannot be used to write loops or conditional statements

D) It cannot interface with input/output devices

Answer: (B)

Explanation: *Since machine language is hardware-specific, a program written for a 32-bit Intel CPU will not run on an ARM processor. Every CPU change requires complete rewriting of the program — unlike high-level languages.*

Chapter 8: Assembly Language

Q90. Assembly language is best defined as:

A) A high-level language that abstracts all hardware details

B) A low-level programming language that uses symbolic mnemonics to represent machine language instructions

C) A language used exclusively for web development

D) A machine-independent language understood by all processors

Answer: (B)

Explanation: *Assembly language replaces binary machine codes with human-readable mnemonics (like MOV, ADD, SUB). It still corresponds closely to machine language but is far easier for programmers to read and write.*

Q91. Which of the following is an example of an assembly language mnemonic?

A) print('Hello')

B) SELECT * FROM table

C) MOV AX, BX

D)

Answer: (C)

Explanation: *MOV AX, BX is a typical x86 assembly instruction meaning 'copy the value in register BX into register AX'. print() is Python, SELECT is SQL, and is HTML markup.*

Q92. The program that translates assembly language code into machine language is called an:

A) Interpreter

B) Compiler

C) Assembler

D) Linker

Answer: (C)

Explanation: *An assembler converts assembly language source code into machine code (object code). It performs a near one-to-one translation since each assembly instruction maps to one machine language instruction.*

Q93. Assembly language is classified as a:

A) First generation language (1GL)

B) Second generation language (2GL)

C) Third generation language (3GL)

D) Fourth generation language (4GL)

Answer: (B)

Explanation: *Assembly language is 2GL because it was the second level of abstraction above raw binary. It replaced binary codes with mnemonics, making programming somewhat easier while still being hardware-dependent.*

Q94. Which of the following is an advantage of assembly language over machine language?

A) Assembly language programs run slower than machine language programs

B) Assembly language is easier to read and write because it uses meaningful mnemonics instead of binary codes

C) Assembly language is machine-independent and portable across all platforms

D) Assembly language does not require any translation before execution

Answer: (B)

Explanation: *Mnemonics like ADD, SUB, LOAD are far more memorable than binary patterns. Assembly programs are also somewhat easier to debug than pure binary, while still allowing direct hardware manipulation.*

Q95. Which of the following is a disadvantage of assembly language?

A) It cannot be used to control hardware directly

B) It is machine-dependent; programs written for one CPU type cannot run on a different CPU without modification

C) It is always slower than high-level language programs

D) It cannot perform arithmetic operations

Answer: (B)

Explanation: *Like machine language, assembly code is processor-specific. An x86 assembly program requires rewriting to run on an ARM processor. This lack of portability is its main disadvantage compared to high-level languages.*

Q96. Assembly language is still used today primarily for:

A) Developing large-scale enterprise web applications

B) Writing device drivers, firmware, and performance-critical code that requires direct hardware access

C) Creating graphical user interfaces and desktop applications

D) Building relational database systems

Answer: (B)

Explanation: *Assembly is used where direct hardware control and maximum performance matter: OS kernel routines, bootloaders, device drivers, real-time systems, and cryptographic implementations that must run at hardware speed.*

Q97. Which of the following correctly describes a mnemonic in assembly language?

A) A binary string that the CPU executes directly

B) An abbreviated symbolic name representing a machine language instruction (e.g., ADD, SUB, MOV)

C) A high-level language keyword like 'while' or 'for'

D) A hexadecimal address in memory

Answer: (B)

Explanation: *Mnemonics are abbreviated memory aids. In assembly, ADD means addition, SUB means subtraction, MOV means move data. Each mnemonic corresponds to a specific binary opcode in the CPU's instruction set.*

Q98. What is a label in assembly language?

A) A binary code for a specific hardware register

B) A symbolic name given to a memory address or instruction location in the code

C) A comment added by the programmer for documentation

D) An instruction that moves data between registers

Answer: (B)

Explanation: *Labels in assembly give human-readable names to memory addresses. For example, 'LOOP_START:' marks the beginning of a loop so the programmer can jump to it by name rather than specifying a raw memory address.*

Q99. Which of the following describes the relationship between assembly language and machine language?

A) They are entirely different languages with no relation to each other

B) Assembly language is a symbolic representation of machine language; there is a near one-to-one correspondence

C) Machine language is a high-level abstraction of assembly language

D) Assembly language is machine-independent while machine language is not

Answer: (B)

Explanation: *Each assembly mnemonic corresponds to exactly one machine language instruction. The assembler simply replaces symbolic names with their binary equivalents, so the relationship is direct and one-to-one.*

Q100. Assembly language programs generally execute:

A) Slower than programs written in Python or Java

B) At roughly the same speed as machine language since the assembler translates directly to machine code

C) Much slower due to the assembler's translation overhead

D) Only on computers with specialized co-processors

Answer: (B)

Explanation: *After assembly (translation), the resulting machine code runs at native CPU speed. The translation overhead happens once at assembly time, not at runtime, so execution speed equals machine language performance.*

Q101. Which of the following is NOT a typical use of assembly language?

A) Writing operating system kernels

B) Programming microcontrollers

C) Developing social media web applications

D) Writing performance-critical routines in compilers

Answer: (C)

Explanation: *Social media web applications are large, complex projects requiring rapid development — they are written in high-level languages (JavaScript, Python, Ruby). Assembly is reserved for low-level, hardware-close programming.*

Q102. The process of converting assembly language source code into machine code is called:

A) Compilation

B) Interpretation

C) Assembly

D) Linking

Answer: (C)

Explanation: *The process performed by an assembler is called 'assembly'. Compilation is for high-level languages; interpretation executes source code line-by-line; linking combines object files into an executable.*

Chapter 9: High-Level Languages

Q103. High-level programming languages are characterized by:

A) Using binary codes and hardware-specific instructions

B) Using English-like syntax and being independent of specific hardware architecture
C) Requiring no translation to run on a CPU
D) Being limited to mathematical calculations only

Answer: (B)

Explanation: *High-level languages use syntax similar to human language (e.g., if, while, print), hiding hardware complexity. They are platform-independent — the same source code can run on different hardware with a compatible compiler.*

Q104. Which of the following is an example of a high-level programming language?

A) Assembly Language
B) Machine Language
C) Python
D) BIOS instruction set

Answer: (C)

Explanation: *Python is a modern high-level language known for its readable syntax. Assembly and machine language are low-level. BIOS instructions are firmware-level code specific to hardware initialisation.*

Q105. High-level languages belong to which generation of programming languages?

A) First generation (1GL)
B) Second generation (2GL)
C) Third generation (3GL) and above
D) Zero generation (0GL)

Answer: (C)

Explanation: *3GLs (like C, Java, Python, Pascal) were the first to use structured English-like syntax and support complex data structures. 4GLs and 5GLs are even higher-level, more domain-specific abstractions.*

Q106. A compiler translates high-level language code by:

A) Translating and executing the program one line at a time
B) Translating the entire source code into machine code before execution
C) Running the program in a virtual environment without translation
D) Converting the code to assembly language only

Answer: (B)

Explanation: *A compiler scans the entire source code, checks for errors, optimises, and produces a complete machine code executable. Once compiled, the executable runs independently without the compiler being present.*

Q107. An interpreter translates high-level language code by:

A) Converting the entire source code to machine code before running
B) Translating and executing one statement at a time during runtime
C) Producing an executable file that can be run independently
D) Assembling the code into binary object files

Answer: (B)

Explanation: *An interpreter reads one statement, translates it, executes it, then moves to the next. Python uses an interpreter. This means the source code must be present every time the program runs.*

Q108. Which of the following is an advantage of compiled languages over interpreted ones?

A) Errors are easier to find since code runs line by line
B) Compiled programs generally run faster because translation happens before execution

C) Compilation does not require the source code to be complete before running

D) Compiled code can be run on any machine without modification

Answer: (B)

Explanation: *Compiled code runs faster because translation overhead is eliminated at runtime. The machine code executable runs directly on the CPU without any intermediate translation step during execution.*

Q109. Which of the following is an advantage of interpreted languages?

A) They produce faster executables than compiled languages

B) They are easier to debug since errors are reported line by line during execution

C) They require no programming knowledge to use

D) Interpreted code is more portable as machine code than compiled code

Answer: (B)

Explanation: *When an interpreter encounters an error, it reports it immediately at that line, making it easier to locate and fix bugs interactively. This is useful during development and learning.*

Q110. 'Portability' as an advantage of high-level languages means:

A) High-level code can be easily moved to a USB drive

B) Programs can run on different types of computers with minimal or no modification

C) Programs are smaller in file size than machine language programs

D) Programs written in high-level languages execute faster

Answer: (B)

Explanation: *The same Python or Java source code can run on Windows, Linux, and macOS with little or no modification. This platform-independence is a major advantage over machine and assembly language.*

Q111. Which of the following is NOT a high-level programming language?

A) C++

B) Java

C) FORTRAN

D) Assembly (x86)

Answer: (D)

Explanation: *x86 Assembly is a low-level 2nd generation language closely tied to the Intel/AMD hardware architecture. C++, Java, and FORTRAN are all high-level 3GL languages with structured, hardware-independent syntax.*

Q112. Object-Oriented Programming (OOP) is a paradigm supported by high-level languages where programs are organized around:

A) Sequential binary instructions

B) Objects that encapsulate data and behaviour

C) Low-level memory addresses

D) Boolean logic gates

Answer: (B)

Explanation: *OOP models real-world entities as objects. Each object bundles data (attributes) and functions (methods). Key OOP principles: encapsulation, inheritance, polymorphism, and abstraction. Java and Python are OOP languages.*

Q113. FORTRAN was primarily developed for:

A) Business data processing

B) Artificial intelligence research

C) Scientific and mathematical computations

D) Web application development

Answer: (C)

Explanation: *FORTRAN (FORmula TRANslation, 1957) was the first widely used high-level language, designed by IBM for scientists and engineers to write complex mathematical formulas efficiently. It is still used in scientific computing today.*

Q114. COBOL was primarily designed for:

A) Scientific calculations and simulations

B) Business and administrative data processing

C) Systems programming and OS development

D) Embedded systems and firmware

Answer: (B)

Explanation: *COBOL (Common Business-Oriented Language, 1959) was designed for business applications: payroll, inventory, banking. Its English-like syntax made it accessible to non-mathematical business users.*

Q115. Which high-level language is considered the most widely used for general-purpose programming today and is known for its simple, readable syntax?

A) COBOL

B) FORTRAN

C) Python

D) Pascal

Answer: (C)

Explanation: *Python consistently ranks as the world's most popular language. Its clean, indentation-based syntax, vast library ecosystem, and versatility (AI, web, data science, automation) make it suitable for beginners and experts alike.*

Q116. A program written in a high-level language is called:

A) Object code

B) Machine code

C) Source code

D) Firmware

Answer: (C)

Explanation: *Source code is the human-readable text written by the programmer in a high-level language. It must be translated by a compiler or interpreter into machine code before the CPU can execute it.*

Q117. After compilation, the translated machine language version of the program is called:

A) Source code

B) Object code / executable code

C) Pseudocode

D) Assembly code

Answer: (B)

Explanation: *After a compiler processes source code, it produces object code (machine code). This may be linked with library code to produce a final executable file (.exe on Windows, ELF on Linux).*

Q118. Which of the following languages is primarily used for developing Android applications?

A) COBOL

B) Pascal

C) Java / Kotlin

D) FORTRAN

Answer: (C)

Explanation: *Google officially supports Java and Kotlin for Android app development. Kotlin is now the preferred language. Both run on the Android Runtime (ART) and compile to bytecode compatible with Android devices.*

Q119. 4GL (Fourth Generation Languages) are characterized by:

A) Being closer to machine language than 3GL

B) Being non-procedural and allowing users to specify WHAT they want without specifying HOW

C) Requiring knowledge of binary to use effectively

D) Being used only in embedded systems

Answer: (B)

Explanation: *4GLs are closer to human thought — the programmer states what result is needed, not how to achieve it step by step. SQL (query databases), report generators, and spreadsheet formula languages are 4GL examples.*

Q120. SQL (Structured Query Language) is an example of which generation language?

A) 1GL

B) 2GL

C) 3GL

D) 4GL

Answer: (D)

Explanation: *SQL is a 4GL because it is declarative — you write SELECT name FROM students WHERE grade > 80 and the database engine works out HOW to retrieve that data. You specify the goal, not the steps.*

Chapter 10: Stored Program Concept

Q121. The Stored Program Concept states that:

A) Programs must be stored on external devices and loaded each time

B) Both program instructions and data are stored in the computer's memory and can be executed sequentially

C) Programs are hardwired into the CPU and cannot be changed

D) Only data is stored in memory; instructions are hardcoded in the hardware

Answer: (B)

Explanation: *The Stored Program Concept (1945) revolutionised computing by allowing instructions and data to coexist in the same memory. The CPU can then fetch instructions from memory, decode them, and execute them sequentially.*

Q122. The Stored Program Concept is most closely associated with which computer architecture?

A) Harvard Architecture

B) Von Neumann Architecture

C) RISC Architecture

D) Pipeline Architecture

Answer: (B)

Explanation: *John von Neumann's architecture stores both program instructions and data in a single memory unit, with a single bus connecting memory to the CPU. Most modern personal computers are based on this architecture.*

Q123. Who is credited with developing the foundational ideas of the Stored Program Concept?

A) Charles Babbage
B) Alan Turing and John von Neumann
C) Bill Gates
D) Tim Berners-Lee

Answer: (B)

Explanation: *Alan Turing's theoretical work on the Universal Turing Machine and John von Neumann's 1945 First Draft of a Report on EDVAC together established the theoretical and practical foundation of the stored program computer.*

Q124. In the Von Neumann architecture, program instructions and data are stored in:

A) Separate memory units to prevent conflicts
B) The same memory unit
C) The CPU registers only
D) External storage devices exclusively

Answer: (B)

Explanation: *The Von Neumann model uses a single unified memory for both code and data. This simplifies design but creates the Von Neumann bottleneck since both compete for the same memory bus bandwidth.*

Q125. The Von Neumann bottleneck refers to:

A) The CPU running out of registers during complex calculations
B) The limited speed of data transfer between the CPU and shared memory causing performance constraints
C) The inability of the computer to run more than one program
D) The limited storage capacity of ROM chips

Answer: (B)

Explanation: *Since a single bus carries both instructions and data, the CPU often has to wait for memory access. Modern systems mitigate this with caches, pipelines, and wider buses, but the fundamental constraint remains.*

Q126. Which of the following is a key advantage of the Stored Program Concept?

A) Programs are burned into hardware and cannot be modified
B) The same computer can run different programs without physical rewiring
C) Only one type of program can be stored in memory at a time
D) Data and instructions are processed at different speeds

Answer: (B)

Explanation: *Before stored programs, computers had to be physically rewired to run different programs. With the stored program concept, swapping programs is as simple as loading different instructions into memory — enabling general-purpose computing.*

Q127. The Fetch-Decode-Execute cycle is central to the Stored Program Concept. In this cycle, what happens during the FETCH stage?

A) The CPU performs the arithmetic or logical operation
B) The CPU retrieves the next instruction from memory

C) The CPU converts the binary instruction into a control signal

D) The CPU writes the result back to memory

Answer: (B)

Explanation: *During FETCH, the CPU uses the Program Counter to find the address of the next instruction, places that address in the MAR, and copies the instruction from memory into the Memory Data Register and then the Current Instruction Register.*

Q128. During the DECODE stage of the Fetch-Decode-Execute cycle:

A) The instruction is fetched from memory

B) The CPU interprets (decodes) the fetched instruction to determine what action is needed

C) The arithmetic logic unit (ALU) performs the calculation

D) The result is stored back in memory or a register

Answer: (B)

Explanation: *In the DECODE stage, the Control Unit analyses the binary instruction to determine what operation is required (ADD, LOAD, JUMP, etc.) and which operands are involved, then signals the appropriate hardware units.*

Q129. The Program Counter (PC) in a CPU:

A) Counts the number of programs installed on the computer

B) Holds the memory address of the next instruction to be fetched and executed

C) Stores the result of the most recent arithmetic operation

D) Controls the speed of the CPU clock

Answer: (B)

Explanation: *The Program Counter is automatically incremented after each fetch to point to the next instruction. Branch and jump instructions modify the PC to change program flow, enabling loops and conditional execution.*

Q130. The Memory Address Register (MAR) is used to:

A) Store the instruction currently being decoded

B) Hold the address in memory from which data or an instruction is to be fetched

C) Temporarily hold the data that has just been read from memory

D) Determine the next instruction based on a condition

Answer: (B)

Explanation: *The MAR is connected to the address bus. Before any memory read or write, the CPU loads the target address into the MAR. The memory system then accesses that specific location.*

Q131. The Memory Data Register (MDR) is used to:

A) Hold the address of the memory location being accessed

B) Temporarily hold the data that is being transferred to or from memory

C) Count the number of executed instructions

D) Store permanent program files on the hard disk

Answer: (B)

Explanation: *The MDR (also called Memory Buffer Register) acts as a buffer between the CPU and memory. Data read from memory is first placed in the MDR before being processed; data to be written is placed here before being sent to memory.*

Q132. Harvard Architecture differs from Von Neumann Architecture in that it:

A) Uses a single memory for both programs and data

B) Uses separate memory and buses for instructions and data, allowing simultaneous access

C) Does not support the Stored Program Concept

D) Only works with assembly language programs

Answer: (B)

Explanation: *Harvard Architecture uses physically separate storage and signal pathways for instructions and data. This eliminates the Von Neumann bottleneck, allowing the CPU to fetch the next instruction while processing data simultaneously. Used in many microcontrollers and DSPs.*

Q133. Which of the following is a practical benefit of the Stored Program Concept in modern computing?

A) It eliminates the need for an operating system

B) Programs can be easily updated, replaced, or modified without any hardware changes

C) The computer can only run one fixed program permanently

D) Data and instructions are completely separate, speeding up processing

Answer: (B)

Explanation: *Because programs are data stored in memory, they can be loaded, updated, patched, or replaced simply by writing new bytes to memory — no hardware changes needed. This flexibility underpins all modern software distribution and updating.*

Chapter 11: MS Office Packages

Q134. MS Office is best described as:

A) An operating system developed by Microsoft

B) A suite of productivity application software including word processing, spreadsheet, and presentation tools

C) A firmware package embedded in Microsoft hardware

D) A programming language developed by Microsoft

Answer: (B)

Explanation: *Microsoft Office is a collection of productivity application software. Current MS 365 includes Word, Excel, PowerPoint, Outlook, OneNote, Access, Teams, and more — all designed to enhance workplace and personal productivity.*

Q135. Microsoft Word is primarily used for:

A) Performing complex statistical data analysis

B) Creating, editing, and formatting text documents

C) Building and managing relational databases

D) Designing and delivering slide-based presentations

Answer: (B)

Explanation: *MS Word is a full-featured word processor for creating letters, reports, essays, and books. It supports text formatting, tables, images, mail merge, spelling/grammar checking, and collaborative editing.*

Q136. Microsoft Excel is primarily used for:

A) Writing and formatting letters and reports

B) Creating and managing databases of customer information

C) Organizing data in rows and columns, performing calculations, and creating charts

D) Creating slide presentations for meetings

Answer: (C)

Explanation: *Excel is a powerful spreadsheet application. It organises data in cells (rows × columns), supports hundreds of built-in functions, enables chart creation, pivot tables, conditional formatting, and automation through macros.*

Q137. Microsoft PowerPoint is primarily used for:

A) Performing calculations using formulas and functions
B) Creating slide-show presentations with text, images, and multimedia
C) Writing, editing, and formatting long text documents
D) Managing and querying relational databases

Answer: (B)

Explanation: *PowerPoint is presentation software. Users create a sequence of slides containing text, images, charts, animations, and videos. Presentations are used in classrooms, boardrooms, and conferences.*

Q138. Microsoft Access is primarily used for:

A) Creating slide presentations
B) Word processing and document formatting
C) Building and managing relational databases
D) Creating spreadsheets and performing data analysis

Answer: (C)

Explanation: *MS Access is a relational database management system (RDBMS). Users can create tables, define relationships, build forms and reports, and write queries using QBE (Query By Example) or SQL.*

Q139. Microsoft Outlook is primarily used for:

A) Creating financial models and spreadsheets
B) Managing emails, contacts, calendar events, and tasks
C) Designing corporate logos and branding materials
D) Editing audio and video files

Answer: (B)

Explanation: *Outlook is a personal information manager and email client. It integrates email, a calendar for scheduling meetings, a contact directory, task management, and notes — all in one application.*

Q140. In Microsoft Word, the feature that automatically corrects common spelling mistakes as you type is called:

A) AutoSave
B) AutoComplete
C) AutoCorrect
D) SpellCheck

Answer: (C)

Explanation: *AutoCorrect automatically replaces common typos (e.g., 'teh' → 'the') as you type. AutoComplete suggests completions for frequently used phrases. SpellCheck is a manual or background checker, not an auto-replacement tool.*

Q141. In Microsoft Excel, a formula always begins with:

A) A hashtag (#)
B) An equals sign (=)
C) A dollar sign ($)
D) An ampersand (&)

Answer: (B)

Explanation: *The equals sign (=) tells Excel that what follows is a formula to be calculated, not plain text. For example, =A1+B1 adds the values in cells A1 and B1. Without =, Excel treats the entry as literal text.*

Q142. What is a 'cell' in Microsoft Excel?

A) A type of chart used to display data
B) The intersection of a row and a column in a worksheet, identified by a letter and number (e.g., A1)
C) A range of data selected for a formula
D) A special macro used to automate tasks

Answer: (B)

Explanation: *A cell is the basic unit of a spreadsheet. Column letter + row number = cell reference (e.g., C5 = column C, row 5). Each cell can hold text, numbers, dates, or formulas.*

Q143. In Excel, a 'workbook' refers to:

A) A single row of data in a spreadsheet
B) The entire Excel file, which can contain multiple worksheets
C) A named range of cells used in a formula
D) A chart embedded within a worksheet

Answer: (B)

Explanation: *A workbook is the entire .xlsx file. It can contain multiple worksheets (tabs) like Sheet1, Sheet2, etc. This allows related data to be organised in separate sheets within a single file.*

Q144. Which of the following is a function in Microsoft Excel that adds up a range of numbers?

A) =COUNT(A1:A10)
B) =AVERAGE(A1:A10)
C) =SUM(A1:A10)
D) =MAX(A1:A10)

Answer: (C)

Explanation: *=SUM(A1:A10) adds all numeric values in cells A1 through A10. COUNT counts how many cells have numbers; AVERAGE calculates the mean; MAX returns the largest value.*

Q145. In Microsoft PowerPoint, a 'slide transition' refers to:

A) The animation applied to individual text or image objects within a slide
B) The visual effect that occurs when moving from one slide to the next during a presentation
C) The template theme applied to all slides
D) The process of converting a presentation to PDF format

Answer: (B)

Explanation: *Slide transitions are effects (fade, wipe, fly-in) that play between slides during a slideshow. They differ from animations, which apply to individual objects within a single slide.*

Q146. Which of the following file extensions is associated with Microsoft Word documents?

A) .xlsx
B) .pptx
C) .docx
D) .accdb

Answer: (C)

Explanation: *.docx is the Word document format (introduced in Office 2007, based on Open XML). .xlsx = Excel, .pptx = PowerPoint, .accdb = Access database.*

Q147. Which of the following file extensions is associated with Microsoft Excel files?

A) .docx
B) .xlsx
C) .pptx
D) .mdb

Answer: (B)

Explanation: *.xlsx stands for Excel Spreadsheet (XML-based format since Office 2007). The older format was .xls. .docx is Word, .pptx is PowerPoint, .mdb was the older Access format (replaced by .accdb).*

Q148. Which of the following file extensions is associated with Microsoft PowerPoint files?

A) .pptx
B) .docx
C) .xlsx
D) .accdb

Answer: (A)

Explanation: *.pptx is the PowerPoint Presentation format. A .ppsx file is a PowerPoint Show (opens directly in slideshow mode). The older format was .ppt. Remember: docx=Word, xlsx=Excel, pptx=PowerPoint.*

Q149. In MS Word, 'Mail Merge' is a feature used to:

A) Send and receive emails directly within Word
B) Combine a document template with a data source to produce personalised letters or labels
C) Merge two separate Word documents into one
D) Format text by merging multiple font styles

Answer: (B)

Explanation: *Mail Merge takes a template letter and a data source (e.g., Excel spreadsheet of names/addresses) and automatically generates personalised copies — useful for sending the same letter to hundreds of people with individual names.*

Q150. Which of the following is NOT a chart type available in Microsoft Excel?

A) Bar Chart
B) Pie Chart
C) Scatter Chart
D) Flow Chart

Answer: (D)

Explanation: *Excel offers Bar, Column, Line, Pie, Area, Scatter, Radar, and many other charts for data visualisation. A Flowchart is a process diagram drawn with drawing tools (or Visio) — it is not a data chart type in Excel.*

Q151. The 'Ribbon' in MS Office applications refers to:

A) A decorative border added to documents
B) The toolbar at the top of the screen that contains organized groups of commands and features
C) The footer area at the bottom of each page
D) The scrollbar on the right side of the document

Answer: (B)

Explanation: *Introduced in Office 2007, the Ribbon replaced traditional menus with a tabbed toolbar. Tabs (Home, Insert, Layout, etc.) contain related command groups, making features more discoverable and accessible.*

Q152. In Microsoft Word, 'Track Changes' is used to:

A) Monitor how long it takes to open a document
B) Automatically save a document at timed intervals
C) Record and display edits made to a document so reviewers can accept or reject them
D) Track the word count as the document grows

Answer: (C)

Explanation: *Track Changes marks every insertion, deletion, and formatting change in a distinctive colour attributed to each reviewer. Document owners can then review changes and choose to accept or reject each one — essential for collaborative editing.*

Q153. Microsoft OneNote is used for:

A) Creating slide presentations for business meetings
B) Taking and organizing notes in a digital notebook format
C) Building relational databases and queries
D) Managing and scheduling emails and calendar events

Answer: (B)

Explanation: *OneNote is a digital notebook application. Users can create notebooks with sections and pages, add text, images, drawings, audio, and hyperlinks. It syncs across devices and supports collaborative note-taking.*

Q154. Which MS Office application would be most suitable for creating a budget tracker?

A) Microsoft Word
B) Microsoft PowerPoint
C) Microsoft Excel
D) Microsoft Access

Answer: (C)

Explanation: *A budget tracker requires numerical data entry, arithmetic formulas, and charts — all strengths of Excel. Word is for text documents, PowerPoint for presentations, and Access for relational databases requiring multiple linked tables.*

Q155. In MS Excel, the VLOOKUP function is used to:

A) Look for a value in the first column of a table and return a value in the same row from a specified column
B) Count the number of cells that contain numerical values
C) Calculate the average of a range of cells
D) Create a vertical bar chart from selected data

Answer: (A)

Explanation: *VLOOKUP (Vertical Lookup) searches for a value in the leftmost column of a table, then returns a value from a specified column in the same row. Syntax: =VLOOKUP(lookup_value, table_array, col_index, [range_lookup]).*

Q156. 'Macro' in MS Office refers to:

A) A large-size document format
B) A set of recorded commands and instructions that can be played back to automate repetitive tasks
C) A type of chart that shows trends over time
D) A password-protected section of a document

Answer: (B)

Explanation: *A macro is a sequence of actions recorded and saved so they can be replayed automatically. In Excel/Word, macros can be recorded manually or written in VBA (Visual Basic for Applications) for complex automation.*

Q157. Which of the following is a key advantage of using MS Office Packages?

A) They replace the need for an operating system

B) They provide a familiar, integrated suite of productivity tools that improve efficiency

C) They are exclusively available for Linux operating systems

D) They can directly compile and run software programs

Answer: (B)

Explanation: *MS Office's greatest advantage is integration: data flows easily between Word, Excel, and PowerPoint. The consistent interface across applications reduces the learning curve, and cloud integration (OneDrive) enables real-time collaboration.*

Consolidated Answer Key

Q1–(B)	Q2–(B)	Q3–(B)	Q4–(B)	Q5–(D)
Q6–(C)	Q7–(B)	Q8–(B)	Q9–(C)	Q10–(C)
Q11–(B)	Q12–(C)	Q13–(B)	Q14–(B)	Q15–(C)
Q16–(C)	Q17–(B)	Q18–(C)	Q19–(B)	Q20–(B)
Q21–(B)	Q22–(C)	Q23–(B)	Q24–(C)	Q25–(B)
Q26–(B)	Q27–(B)	Q28–(B)	Q29–(C)	Q30–(C)
Q31–(B)	Q32–(B)	Q33–(B)	Q34–(C)	Q35–(B)
Q36–(B)	Q37–(C)	Q38–(B)	Q39–(D)	Q40–(B)
Q41–(B)	Q42–(C)	Q43–(B)	Q44–(C)	Q45–(B)
Q46–(B)	Q47–(B)	Q48–(B)	Q49–(B)	Q50–(B)
Q51–(B)	Q52–(C)	Q53–(B)	Q54–(B)	Q55–(B)
Q56–(C)	Q57–(B)	Q58–(C)	Q59–(B)	Q60–(B)
Q61–(B)	Q62–(B)	Q63–(C)	Q64–(B)	Q65–(B)
Q66–(C)	Q67–(B)	Q68–(B)	Q69–(C)	Q70–(B)
Q71–(B)	Q72–(B)	Q73–(B)	Q74–(B)	Q75–(B)
Q76–(B)	Q77–(B)	Q78–(B)	Q79–(B)	Q80–(C)
Q81–(B)	Q82–(B)	Q83–(B)	Q84–(B)	Q85–(C)
Q86–(C)	Q87–(B)	Q88–(D)	Q89–(B)	Q90–(B)
Q91–(C)	Q92–(C)	Q93–(B)	Q94–(B)	Q95–(B)
Q96–(B)	Q97–(B)	Q98–(B)	Q99–(B)	Q100–(B)
Q101–(C)	Q102–(C)	Q103–(B)	Q104–(C)	Q105–(C)
Q106–(B)	Q107–(B)	Q108–(B)	Q109–(B)	Q110–(B)
Q111–(D)	Q112–(B)	Q113–(C)	Q114–(B)	Q115–(C)
Q116–(C)	Q117–(B)	Q118–(C)	Q119–(B)	Q120–(D)
Q121–(B)	Q122–(B)	Q123–(B)	Q124–(B)	Q125–(B)
Q126–(B)	Q127–(B)	Q128–(B)	Q129–(B)	Q130–(B)
Q131–(B)	Q132–(B)	Q133–(B)	Q134–(B)	Q135–(B)
Q136–(C)	Q137–(B)	Q138–(C)	Q139–(B)	Q140–(C)
Q141–(B)	Q142–(B)	Q143–(B)	Q144–(C)	Q145–(B)
Q146–(C)	Q147–(B)	Q148–(A)	Q149–(B)	Q150–(D)
Q151–(B)	Q152–(C)	Q153–(B)	Q154–(C)	Q155–(A)
Q156–(B)	Q157–(B)			